

A disproof of Conjecture 00000004053

The trivial $\mathbb{F}_3[C_3]$ -module has syzygy period 2, and 2 divides no prime factor of the order of any subgroup of C_3

1 The conjecture

Conjecture 00000004053 reads (English):

Definition: A periodic module is one with $\Omega^n M$ isomorphic to M . *Conjecture:* The prime factors of the minimal period of a periodic module always divide prime factors of the order of the defect group, and every power of every admissible prime is realized in some explicit block. (period-defect group prime factor correspondence)

The Chinese text says the same: the prime factors of the minimal period of a periodic module always divide prime factors of the order of the defect group, and every power of every admissible prime is realized in some explicit block.

Reading. The setting is modular representation theory: a finite group G , a field k of characteristic p , a block of kG with its defect group D , and a periodic module M in that block, i.e. a finitely generated module with $\Omega^n M \cong M$ for some $n \geq 1$, where ΩM is the kernel of a projective cover of M (the Heller operator). The *minimal period* is the least such n . Clause 1 says: every prime factor q of the minimal period divides some prime factor r of $|D|$. Since q and r are primes, this is the same as $q = r$, and the same as $q \mid |D|$; we refute both formulations. Clause 2 (“admissible prime”, “explicit block”) is not defined in the text and is not used: the conjecture is a conjunction, and we refute Clause 1.

Counterexample. $p = 3$, $k = \mathbb{F}_3$, $G = C_3$, $M = k$ the trivial module. Its minimal period is 2, while every subgroup of C_3 (in particular every possible defect group) has order 1 or 3; 2 divides neither 3 nor any prime factor of 1 or 3.

2 Definitions (as in the Lean file)

All modules are left modules over a ring R ; in Lean, modules are objects of `ModuleCat.{0} R`.

- A submodule $K \leq P$ is *superfluous* if $K + L = P$ implies $L = P$ (`IsSuperfluous`).
- $\pi : P \rightarrow M$ is a *projective cover* if P is projective, π is onto, and $\ker \pi$ is superfluous (`IsProjectiveCover`). This is the definition retrieved from Wikipedia (*Projective cover*): an epimorphism $p : P \rightarrow M$ “such that the kernel of p is a superfluous submodule of P ”.
- N is a *syzygy* of M (`IsSyzygy R M N`) if $N \cong \ker \pi$ for some projective cover $\pi : P \rightarrow M$. `IterSyz R n M N` means $N \cong \Omega^n M$: there is a chain $M = M_0, M_1, \dots, M_n$ with M_{i+1} a syzygy of M_i and $M_n \cong N$.

- **HasPeriod R M n**: $n \geq 1$, $\Omega^n M \cong M$, and $\Omega^m M \not\cong M$ for $0 < m < n$ (for every chain). So n is the minimal period.
- M is *indecomposable* (**IsIndecomposable**) if $M \neq 0$ and whenever $M = S \oplus T$ for submodules S, T (**IsComp1 S T**), $S = 0$ or $T = 0$.
- Defect group. Wikipedia (*Modular representation theory*): “To each block B of the group algebra $K[G]$, Brauer associated a certain p -subgroup, known as its defect group (where p is the characteristic of K).” We use only that a defect group is a subgroup of G . (Wikipedia also states that “the Sylow p -subgroup of the finite group G is a defect group for the principal block of $K[G]$ ”; for $G = C_3$, $p = 3$ this is C_3 , but we do not need it.)

The group algebra. $G = C_3$ is Mathlib’s **Multiplicative (ZMod 3)** with generator g ; $A = \mathbb{F}_3[C_3]$ is Mathlib’s group algebra **MonoidAlgebra (ZMod 3) G**. Let $t = g - 1 \in A$ and let $\varepsilon : A \rightarrow \mathbb{F}_3$ be the augmentation ($\varepsilon(h) = 1$ for $h \in G$, Mathlib’s **MonoidAlgebra.lift** of the trivial character). The trivial module is $k = \mathbb{F}_3$ with $a \cdot x = \varepsilon(a)x$ (**Triv**); every $h \in G$ acts as the identity (**triv_group_acts_trivially**).

3 Proof

Lemma 1 (**t_cube, t_sq_ne_zero, exists_t_mul**). $t^3 = 0$, $t^2 \neq 0$, and every $a \in A$ is $a = \varepsilon(a) + td$ for some $d \in A$; so $\ker \varepsilon = tA$.

Proof. $(g-1)^3 = g^3 - 1 - 3(g^2 - g) = 0$ since $g^3 = 1$ and $3 = 0$ in A . Also $t^2 = g^2 - 2g + 1 = g^2 + g + 1$, whose coefficient at $1 \in G$ is 1. For the decomposition it suffices (by linearity) to treat $a = h \in G$: $h = g^m$ and $g^m - 1 = (g - 1)(1 + g + \cdots + g^{m-1})$. $\square$

Remark 2. Hence $1, t, t^2$ span A , and since $\dim A = 3$ they form a basis; with $t^3 = 0$ this gives $A \cong \mathbb{F}_3[t]/(t^3)$ via $t = g - 1$. The Lean proof does not use this isomorphism; it works in the group algebra itself, using only the three facts of Lemma 1.

Lemma 3. (a) $t^2c = \varepsilon(c)t^2$ for all c (**t_sq_mul**); hence $t^2c = 0 \Rightarrow \varepsilon(c) = 0$. (b) If $a \neq 0$ then $t^2 \in Aa$ (**t_sq_mem**). (c) Every submodule of A inside $\ker \varepsilon$ is superfluous (**superfluous_of_le_ker**). (d) If P is projective and $t^2p = 0$ then $p \in tP$ (**mem_t_of_t_sq_smul**).

Proof. (a) $c = \varepsilon(c) + td$ and $t^3 = 0$. (b) If $\varepsilon(a) \neq 0$, $t^2 = \varepsilon(a)^{-1}t^2a$. Otherwise $a = td$; if $\varepsilon(d) \neq 0$ then $ta = \varepsilon(d)t^2$; otherwise $a = t^2d' = \varepsilon(d')t^2$ with $\varepsilon(d') \neq 0$. (c) If $K + L = A$ then $1 = x + l$ with $\varepsilon(l) = 1$, $l = 1 + td$, and $(1 - td + t^2d^2)l = 1$, so $L = A$. (d) P is a direct summand of a free module (**Module.projective_def'**); the coordinates c_i of p satisfy $t^2c_i = 0$, so $c_i \in tA$ by (a) and Lemma 1, and $p \in tP$. $\square$

Lemma 4 (cyclic modules). Let $M = Am_0$ with $m_0 \notin tM$ (**CyclicGen**). Then (i) $A \rightarrow M$, $a \mapsto am_0$, is a projective cover (**toSpan_isCover**); (ii) for every projective cover $\pi : P \rightarrow M$, $\ker \pi \cong \text{ann}(m_0)$ (**ker_cover_equiv**).

Proof. (i) If $am_0 = 0$ and $\varepsilon(a) \neq 0$, writing $a = \varepsilon(a) + td$ gives $m_0 = t \cdot (-\varepsilon(a)^{-1}dm_0) \in tM$; so the kernel lies in $\ker \varepsilon$ and is superfluous by Lemma 3(c). (ii) Pick p_0 with $\pi(p_0) = m_0$ and let $\varphi(a) = ap_0$. Then $\text{im } \varphi + \ker \pi = P$, so φ is onto (superfluous kernel). If $ap_0 = 0$ with $a \neq 0$, then $t^2p_0 = 0$ by Lemma 3(b), so $p_0 = tq$ by (d) and $m_0 = t\pi(q)$, a contradiction; so φ is injective. Thus $\varphi : A \cong P$, $\pi \circ \varphi$ is $a \mapsto am_0$, and φ maps $\text{ann}(m_0)$ onto $\ker \pi$. $\square$

Theorem 5 (`period_K`, `iterSyz_two_any`). *The trivial module k is finitely generated, simple (hence indecomposable) and not projective; its minimal syzygy period is 2; and every second syzygy of k , for every choice of projective covers, is isomorphic to k .*

Proof. $k = A \cdot 1$ and $1 \notin tk = 0$. By Lemma 4, $\Omega k \cong N_1 = \text{ann}(1) = \ker \varepsilon$ for every cover. $N_1 = A \cdot t$ and $t \notin tN_1$ (else $t = t^2d$ and $t^2 = t^3d = 0$), so every syzygy of N_1 (or of any module isomorphic to it, `CyclicGen.map`) is isomorphic to $N_2 = \text{ann}(t) = \{a : at = 0\}$. The map $x \mapsto xt^2$ is an A -isomorphism $k \rightarrow N_2$ (`trivEquivN2`): it is A -linear by Lemma 3(a), injective since $t^2 \neq 0$, and onto since $at = 0$ forces $a = t^2d' = \varepsilon(d')t^2$. Finally $N_1 \not\cong k$ since $t \cdot t = t^2 \neq 0$ in N_1 while t kills k (`N1_not_equiv_triv`); so no first syzygy of k is isomorphic to k (`not_iterSyz_one`). Simplicity: any nonzero $x \in k$ generates k (`triv_simple`). Non-projectivity: $t^2 \cdot 1 = 0$ but $1 \notin tk$, contradicting Lemma 3(d) (`triv_not_projective`). $\square$

Theorem 6 (`conjecture_4053_false`, `conjecture_4053_false'`). *Let $\text{Clause1 } p \ H$ say: for every finitely generated, indecomposable, non-projective $\mathbb{F}_p[H]$ -module M with minimal period n there is a subgroup $D \leq H$ such that every prime factor q of n divides some prime factor r of $|D|$ ($\text{Clause1}' : q \mid |D|$). Then $\text{Clause1 } 3 \ C_3$ and $\text{Clause1}' 3 \ C_3$ are false.*

Proof. Take $M = k$, $n = 2$, $q = 2$. By Lagrange (`Subgroup.card_subgroup_dvd_card`), $|D| \mid 3$ for every $D \leq C_3$, so $2 \nmid |D|$ and 2 divides no prime factor of $|D|$ (`not_two_dvd_card`). $\square$

Since the defect group of any block of $\mathbb{F}_3[C_3]$ is a subgroup of C_3 , Clause 1 of the conjecture implies $\text{Clause1 } 3 \ C_3$; hence the conjecture is false (`conjecture_4053_conj_false`: $\neg(\text{Clause1 } 3 \ C_3 \wedge Q)$ for every Q).

4 Relation to the earlier PR #291

PR #291 used the same counterexample ($\mathbb{F}_3[C_3]$, trivial module, period 2, defect group of order 3) and was closed without merging because its Lean file was vacuous: its main theorem was the arithmetic statement $1 = 1 \wedge 2 = 1 + 1 \wedge 3 = 3 \wedge 3 \bmod 2 = 1 \wedge 4 = 2 \cdot 2$, and the review noted that no module, group algebra, Ω -operator or defect group existed in Lean. The present formalization contains the actual objects: Mathlib's group algebra $\mathbb{F}_3[C_3]$, the trivial module with the group acting trivially, projective covers with superfluous kernels, syzygies as kernels of projective covers (for arbitrary projective modules, not a fixed choice), the computation $\Omega k \cong \ker \varepsilon$, $\Omega^2 k \cong k$, $\Omega k \not\cong k$, the minimal-period predicate, and the quantifier over all subgroups of C_3 .

5 Scope

Refuted: Clause 1 (in both formulations above) for $p = 3$, $G = C_3$, $k = \mathbb{F}_3$, with the defect group replaced by an arbitrary subgroup of C_3 , and with periodic modules restricted to finitely generated indecomposable non-projective modules; hence the conjunction. Conventions: syzygies are a relation ($N \cong \ker \pi$ for some projective cover π); we prove that the period does not depend on the choice of covers. Modules in Lean live in `Type` (universe 0); the projective cover P in a syzygy may be any projective module, not only a finitely generated one. Not formalized: blocks and defect groups themselves (only the fact that a defect group is a subgroup of G is used), Clause 2, and the reading in which $\Omega^n M \cong M$ is taken in the stable module category. The same argument works for every odd prime p with $G = C_p$ (not formalized).

6 Lean formalization

File `lean/Conjecture4053/Basic.lean` (namespace `C4053`), only import `Mathlib`, Lean toolchain and `Mathlib` as in `lean-toolchain/lake-manifest.json`. Main statements:

- `conjecture_4053_false : ~Clause1 3 G and conjecture_4053_false' : ~Clause1' 3 G;`
- `trivial_module_counterexample: Module.Finite A Triv, IsIndecomposable A Triv, ~Module.Projective A Triv, HasPeriod A K 2, every IterSyz A 2 K N has N isomorphic to Triv, and for every D : Subgroup G and every prime factor r of Nat.card D, ~2 ∣ r;`
- `conjecture_4053_conj_false (Clause2 : Prop) : ~(Clause1 3 G /\ Clause2).`

Axiom audit (`#print axioms` for all four): `[propext, Classical.choice, Quot.sound]`. No sorry, no `native_decide`, no added axioms. Built with `lake build`.

Sources

- Wikipedia, *Projective cover*, https://en.wikipedia.org/wiki/Projective_cover: “such that the kernel of p is a superfluous submodule of P .”
- Wikipedia, *Modular representation theory*, https://en.wikipedia.org/wiki/Modular_representation_theory: “To each block B of the group algebra $K[G]$, Brauer associated a certain p -subgroup, known as its defect group (where p is the characteristic of K).”; “the Sylow p -subgroup of the finite group G is a defect group for the principal block of $K[G]$.”
- GitHub PR #291 (closed), review comment: <https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/pull/291>.